St. Vincent Pallotti
College of Engineering & Technology

Academic Year: 2023-24

CAT II - B. Tech

Course:	Operating System	**Course Code:**	CB401T
		Max. Marks:	35
Semester:	IV	**Duration:**	1 hr. 30 min
Program:	Computer Science & Business Systems	**Date of Paper**	06-05-2024

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Design a system model that describes the deadlock situation.	02	3	2
Que.1(b)	Define critical section	02	4	2
Que.1(c)	What is logical address and physical address in memory?	02	4	1
Que.1(d)	Define paging.	02	5	1
Que.1(e)	Enlist different file attributes.	02	5	2
Que.2(a)	Consider following snapshot of System:	05	3	3

Process	Allocation	Max	Available
	A B C	A B C	A B C
P0	0 1 0	7 5 3	3 3 2
P1	2 0 0	3 2 2	
P2	3 0 2	9 0 2	
P3	2 1 1	2 2 2	
P4	0 0 2	4 3 3	

Answer the following questions using the banker's algorithm:
a. Calculate Need Matrix?
b. Is the system in a safe state?
c. If a request from process P1 arrives for (1,0,2), can the request be granted immediately?

	OR			
Que.3(a)	Explain Semaphore and it's implementation in detail.	05	3	1
Que.4(a)	Describe in detail about the process of calculating the internal fragmentation.	05	4	1
Que.4(b)	Explain logical and physical address mapping in paging.	05	4	2
	OR			

Que.5(a)	Write a short note on : 1)Need of Memory Management 2)Swapping	05	4
Que.5(b)	Consider the following page reference storing: 1, 2, 3, 4, 3, 2, 5, 1, 3, 2, 1, 5, 2 How many page faults would occur for the following page replacement algorithm. Assuming frame size = 3. i) LRU ii) FIFO	05	4
Que.6(a)	Explain different file operations.	06	5
Que.6(a)	Illustrate different file types.	04	5
	OR		
Que.7(a)	Describe in detail about the directory structure.	05	5
Que.7(b)	Write a short note on: 1) Sequential Access 2) Direct access	05	5

Course Outcomes - After Completion of this course student will -

CO1	Understand the basics of how operating systems work.
CO2	Explain how processes and CPU scheduling function in an operating system.
CO3	Solve common process synchronization problems.
CO4	Describe memory management concepts, including virtual memory.
CO5	Comprehend disk management and the role of file systems in an operating system.

St. Vincent Pallotti
College of Engineering & Technology

Academic Year: 2023-24

CAT I - B. Tech

Course:	**Operating System**	**Course Code:**	CB401T
		Max. Marks:	35
Semester:	**IV**	**Duration:**	1 hr .30 min
Program:	**Computer Science & Business Systems**	**Date**	15/03/24

nstructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of Non-Programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	Define Operating System.	02	1	1
Que.1(b)	Differentiate between User Mode and Kernel mode	02	1	2
Que.1(c)	Identify the basic components of a PCB in an operating system	02	2	1
Que.1(d)	Specify the various types of schedulers.	02	2	1
Que.1(e)	Specify two differentiating points between Network Operating System and Distributed Operating System.	02	3	2
Que.2(a)	Explain the services of Operating System.	05	1	2
Que.2(b)	Provide the significance of System call and it's working principle.	05	1	1
	OR			
Que.3(a)	Illustrate the importance on Batch Operating System.	05	1	2
Que.3(b)	Illustrate the components of Operating System.	05	1	2
Que.4(a)	Elaborate process state model.	05	2	1
Que.4(b)	Differentiate between processes and threads. Provide a diagram illustrating the various states in a thread's life cycle.	05	2	2
	OR			
Que.5(a)	Consider four processes P1, P2, P3, P4 with the length of CPU burst time. Find out the Average Waiting Time and Average Turnaround Time for **FCFS** algorithm (Nonpreemptive)	05	2	3

Process ID	Arrival time	Burst Time
P1	0	2
P2	1	4
P3	3	3
P4	4	6

Que.5	Consider four processes A, B, C, D with the length of CPU burst time. Find out the Average Waiting Time and Average Turnaround Time for following algorithm. 1.Round Robin (Time Quantum =3)	05

Process ID	Arrival time	Burst Time
A	0	5
B	1	7
C	2	9
D	3	4

Que.6	Write short note on User Level and Kernel Level Thread.	05
	OR	
Que.7	Illustrate the process table and PCB.	05

Course Outcomes - After Completion of this course student will be able to -

CO1	Understand the basics of how operating systems work.
CO2	Explain how processes and CPU scheduling function in an operating system.
CO3	Solve common process synchronization problems.
CO4	Describe memory management concepts, including virtual memory.
CO5	Comprehend disk management and the role of file systems in an operating system.

S/24/R

St. Vincent Pallotti College of Engineering & Technology
(An Autonomous Institute)
B.Tech Fourth Semester Computer Science & Business Systems
End Semester Examination

Course: - Operating System **Course Code: -** CB401T

Duration: 3 hours **Max. Marks: -** 70

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) Solve Que. No. 08 OR Que. No. 09
6) Solve Que. No. 10 OR Que. No. 11
7) All Questions carry marks as indicated
8) Use of Non Programmable calculator is allowed.

Q. No.	Description of Question	Mark	CO	BTL
Q.1(a)	Specify the advantages of using Batch operating system	02	1	1
Q.1(b)	Differentiate between Network OS & Distributed OS	02	1	2
Q.1(c)	Identify the basic components of a PCB in an operating system	02	2	1
Q.1(d)	Specify the role of Mid-term scheduler	02	2	1
Q.1(e)	Enlist elements of critical section	02	3	1
Q.1(f)	Elaborate deadlock	02	3	1
Q.1(g)	Write shote note on virtual memory	02	4	2
Q.1(h)	List the advantages of segmentation.	02	4	1
Q.1(i)	Enlist File functions	02	5	2
Q.1(j)	Elaborate Time stamp file attribute	02	5	2
Q.2(a)	Describe the operating system services in detail.	05	1	2
Q.2(b)	Write short note on Device management functions.	05	1	2
Q.3(a)	Specify the types of System call and highlight the functions of each type.	05	1	2
Q.3(b)	Write short note on Information management function.	05	1	1
Q.4(a)	Illustrate the concept and significance of Inter process communication	05	2	3
Q.4(b)	Differentiate between processes and threads.	05	2	2
Q.5(a)	Consider the processes P1, P2, P3, P4 and P5 with the length of CPU burst time. Find out the Average Waiting Time and Average Turnaround Time for FCFS algorithm.	05	2	3
Q.5(b)	Differentiate between Preemptive and Non-Preemptive scheduling algorithm	05	2	3

Process ID	Arrival time	Burst Time
P1	2	6
P2	5	2
P3	1	8
P4	0	3
P5	4	4

Q.6(a)		05

Process ID	Allocation	Max	Available
	A B C D	A B C D	A B C D
T0	0 0 1 2	0 0 1 2	1 5 2 0
T1	1 0 0 0	1 7 5 0	
T2	1 3 5 4	2 3 5 6	
T3	0 6 3 2	0 6 5 2	
T4	0 0 1 4	0 6 5 6	

i) Calculate the NEED matrix
ii) Is the system in a safe state?

Q.6(b)	Explain the Dinning Philospher problem in detail.	05
Q.7(a)	Elaborate MUTEX locks in detail.	05
Q.7(b)	Elaborate Semaphore and its implementation.	05
Q.8(a)	Explain contiguous & non contiguous memory allocation.	05
Q.8(b)	Explain logical to physical address translation in paging.	05
Q.9(a)	Write a short note on: 1) Best Fit 2) First Fit	05
Q.9(b)	Write short note on 1) Trashing 2) Page fault 3) Swapping	05
Q.10(a)	Write a short note on following function 1) Repositioning within a file. 2) Truncating a file.	05
Q.10(b)	Elaborate various file types with its usual extension and function. 1) Word processor 2) Source code 3) Library 4) Multimedia	05
Q.11(a)	Elaborate various file access methods	05
Q.11(b)	Write short note on following file attributes 1) Identifier 2) Location 3) Protection 4) User Identification	05

St. Vincent Pallotti College of Engineering & Technology

(An Autonomous Institute)

Academic Year: 2023-24

CAT I - B.Tech

urse: -	Introduction to Innovation, IP Management & Entrepreneurship	Course Code: -	CB404T
		Max. Marks: -	35
nester: -	IV	Duration: -	1 hr 30 min
gram : -	Computer Science & Business Systems	Date of Paper	19/03/2024

ctions to Candidate –
1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of Non Programmable calculator is allowed.

ue. No.	Description of Question	Marks	[CO]	[BTL]
.1(a)	Define Creativity and explain the term "innovation".	02	1	2
.1(b)	Define Design thinking and List 5 stages of design thinking.	02	1	2
.1(c)	Explain the term "open Innovation" & list any 4 strategies for Fostering Creativity and Experimentation.	02	1	2
.1(d)	Explain the term "traditional or closed innovation" & list any 4 key advantages of open innovation model.	02	1	2
.1(e)	Explain the term "Entrepreneur" & the term " Social Entrepreneur".	02	2	2
.2(a)	Explain Knowledge Push: Innovation and identify at least 4 real life application.	05	1	3
.2(b)	Identify and explain the problem and proposed solution as per respective team.	05	1	3
	OR			
.3(a)	Explain Need Pull: Innovation and identify at least 4 real life applications.	05	1	3
.3(b)	Is innovation manageable or just a random gambling activity? Comment in detail.	05	1	3
.4(a)	Identify and explain any 3 key Characteristics of open innovation model.	05	1	3
.4(b)	Innovation Model : Co-operating across networks(Open Innovation) vs. 'go-it-alone' approach(Traditional or Closed Innovation) Comment in detail.	05	1	3

	OR	
Que.5(a)	Identify & explain the barriers to innovation in entrepreneurship.	05
Que.5(b)	Identify and explain the innovation/uniqueness in the proposed solution as per respective team's identified problem.	05
Que.6	Explain "Entrepreneurship" in detail.	05
	OR	
Que.7	Explain any two internal and external sources of identifying business idea/opportunities.	05

Course Outcomes - After Completion of this course student will be able -

CO1	Learn to be familiar with creative and innovative thinking styles
CO2	Learn to investigate, understand and internalize the process of founding a startup
CO3	Learn to manage various types of IPR to protect competitive advantage

St. Vincent Pallot
College of Engineering & Technolo

Academic Year: 2023-24

CAT II - B.Tech

se: -	Introduction to Innovation, IP Management & Entrepreneurship	Cou Ma
ster: -	IV	Du
ram : -	Computer Science & Business Systems	Da

ions to Candidate – 1) Question No. 1 is compulsory
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of Non Programmable calculator is

No.	Description of Question
(a)	Differentiate between "Business Model" and "Business
(b)	List the key sections of a P&L statement.
(c)	List types of IP (Intellectual Property).
(d)	Identify the type of IPR as per respective team's propo
(e)	List all types of market.
(a)	Explain "Customer acquisition funnel" of sales plan.
(b)	Explain financial plan in detail.
	OR
(a)	State and explain any 4 methods of Business Valua
(b)	Explain the 6 stages of funding in Entrepreneurshi
(a)	Describe the Genesis and Development of IPR in
(b)	Illustrate how IP management intersects with ma
	OR
(a)	Demonstrate the infringement and penalties wi
(b)	Explain the term "Geographical Indications (for protecting them,
	Outline market segmentation and its types.

Academic Year: 2023-24

CAT II - B.Tech

ourse: -	Introduction to Innovation, IP Management & Entrepreneurship	**Course Code: -**	CB4041
		Max. Marks: -	35
emester: -	IV	**Duration: -**	1 hr 30 min
rogram : -	Computer Science & Business Systems	**Date of Paper**	09/05/2024

ructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated
6) Use of Non Programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
)ue.1(a)	Differentiate between "Business Model" and "Business Plan"	02	2	2
)ue.1(b)	List the key sections of a P&L statement.	02	2	1
)ue.1(c)	List types of IP (Intellectual Property).	02	3	1
)ue.1(d)	Identify the type of IPR as per respective team's proposed solution.	02	3	2
)ue.1(e)	List all types of market.	02	2	1
)ue.2(a)	Explain "Customer acquisition funnel" of sales plan.	05	2	2
)ue.2(b)	Explain financial plan in detail.	05	2	2
	OR			
)ue.3(a)	State and explain any 4 methods of Business Valuation.	05	2	3
)ue.3(b)	Explain the 6 stages of funding in Entrepreneurship.	05	2	3
)ue.4(a)	Describe the Genesis and Development of IPR in India.	05	3	3
)ue.4(b)	Illustrate how IP management intersects with marketing	05	3	2
	OR			
)ue.5(a)	Demonstrate the infringement and penalties with respect to patents.	05	3	2
)ue.5(b)	Explain the term "Geographical Indications (GIs)" and give reasons for protecting them,	05	3	3
)ue.6	Outline market segmentation and its types.	05	2	2
	OR			
)ue.7	Explain with neat sketch the lean canvas for respective teams' problem statement.	05	2	2

St. Vincent Pallotti College of Engineering & Technology
(An Autonomous Institute)
B. Tech Fourth Computer Science & Business Systems
End Semester Examination

S|24|R

Course: - Introduction to Innovation, IP Management & Entrepreneurship **Course Code: -** CB404T

Duration: - 3 hours **Max. Marks: -** 70

Instructions to Candidate –
1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) Solve Que. No. 08 OR Que. No. 09
6) Solve Que. No. 10 OR Que. No. 11
7) All Questions carry marks as indicated
8) Use of Non Programmable calculator is allowed.

Q. No.	Description of Question	Mark	CO	BTL
Q.1(a)	Explain the term "Human centered Design".	02	1	2
Q.1(b)	List 5 stages of design thinking.	02	1	2
Q.1(c)	List any 4 key challenges of open innovation model.	02	1	2
Q.1(d)	List any 4 Real life examples of Open Innovation.	02	1	2
Q.1(e)	List any 2 internal and external sources of business ideas.	02	2	2
Q.1(f)	Identify the type of market for respective team's proposed solution.	02	2	2
Q.1(g)	List the key sections of a P&L statement.	02	2	2
Q.1(h)	Explain the term "Business Valuation".	02	2	2
Q.1(i)	Identify the type of IPR as per respective team's proposed solution.	02	3	2
Q.1(j)	List types of IP (Intellectual Property)	02	3	2
Q.2(a)	Explain Knowledge Push: Innovation and identify at least 4 real life applications.	05	1	2
Q.2(b)	Explain Need Pull: Innovation and identify at least 4 real life application.	05	1	2
Q.3(a)	Is innovation manageable or just a random gambling activity? Comment in detail.	05	1	2
Q.3(b)	Identify and explain the problem and proposed solution as per respective team.	05	1	2
Q.4(a)	Identify & explain key Characteristics of open innovation model.	05	1	2
Q.4(b)	Explain the importance of innovation in business success.	05	1	2
Q.5(a)	Innovation Model : Co-operating across networks(Open Innovation) vs. 'go-it-alone' approach(Traditional or Closed Innovation) Comment in detail.	05	1	2
Q.5(b)	Identify and explain the innovation in the proposed solution as per respective team.	05	1	2
Q.6(a)	Explain how Intellectual Property can be leveraged for Competitive Advantage.	05	2	2
Q.6(b)	Explain market segmentation and its types.	05	2	2
Q.7(a)	Explain the concept of "A Good Business Idea".	05	2	2
Q.7(b)	Explain with neat sketch the lean canvas for respective team's problem statement.	05	2	2
Q.8(a)	Explain financial Forecasting/Projections in detail.	05	2	2

Q.8(b) Explain any 4 methods of Business Valuation.

Q.9(a) Explain "Customer acquisition funnel" of sales plan.

Q.9(b) Explain financial plan in detail.

Q.10(a) Explain IPR from the International context.

Q.10(b) Explain how IP management intersects with marketing.

Q.11(a) Explain the economics Behind the Development of IPR.

Q.11(b) Explain the Genesis and Development of IPR in India.

St. Vincent Pallotti
College of Engineering & Technology

Academic Year: 2023-24

CAT I - B.Tech.

rse: -	**Indian Knowledge System**	**Course Code: -**	**ES106T**
		Max. Marks: -	20
ester: -	**II / IV**	**Duration: -**	1 hr
gram: -	**Group – II and CSBS**	**Date of Paper**	20/02/2024

ructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated
5) Use of Non-Programmable calculator is allowed.

ue. No.	Description of Question	Marks	[CO]	[BTL]
ue.1(a)	Why do we need IKS?	01	CO1	BT1
ue.1(b)	What is the goal of Ancient Indian Education?	01	CO1	BT1
ue.1(c)	What are the contributions of Ved Vyas?	01	CO2	BT1
ue.1(d)	Who founded the Vaishesika school of philosophy?	01	CO2	BT1
ue.1(e)	Name the birthplace of Acharya Mahavir.	01	CO2	BT1
ue.2 (a)	Describe prime factors of Ancient Indian Knowledge.	04	CO1	BT2
ue.2 (b)	Explain any four features of IKS.	04	CO1	BT2
	OR			
Que.3 (a)	Discuss the achievements of IKS?	04	CO1	BT2
Que.3 (b)	Give classification framework (IKS corpus) of IKS.	04	CO1	BT2
Que.4 (a)	Discuss Contributions of Acharya Nagarjuna in IKS.	03	CO2	BT2
Que.4 (b)	What are the astronomical discoveries of Maharshi Aryabhatta?	04	CO2	BT1
	OR			
Que.5 (a)	Explain about the importance of Charak Sanhita in Ayurveda	03	CO2	BT2
Que.5 (b)	Enlist four Vedas. What is the main content of four Vedas?	04	CO2	BT1

Course Outcomes - After Completion of this course student will -

CO1	Explain the origin, evolution and approach of Indian knowledge systems
CO2	Describe importance of Indian Literature and will know about Indian Scholars
CO3	Understand ancient Indian traditions, practices and knowledge for conservation of resources
CO4	Understand and build various practices to conserve and protect rich Indian Knowledge

St. Vincent Pallotti
College of Engineering & Technology

Academic Year: 2023-24

CAT II - B.Tech.

Course: -	**Indian Knowledge System**	**Course Code: -**	**ES106T**
		Max. Marks: -	20
Semester: -	**II / IV**	**Duration: -**	1 hr
Program: -	**Group – II and CSBS**	**Date of Paper**	13/05/2024

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated
5) Use of Non-Programmable calculator is allowed.

Que. No.	Description of Question	Marks	[CO]	[BTL]
Que.1(a)	What is the moral of the story "Integrity of Ram"?	01	CO3	BT2
Que.1(b)	What do you mean by Jataka Tales?	01	CO3	BT1
Que.1(c)	What "Ayurveda" means in Sanskrit?	01	CO3	BT1
Que.1(d)	In which year Biodiversity Act was introduced?	01	CO4	BT1
Que.1(e)	What is TKDL?	01	CO4	BT1
Que.2 (a)	Describe Spiritualism in India on the basis of three central values?	04	CO3	BT2
Que.2 (b)	Explain in short about various major traditional medicine systems in India.	04	CO3	BT2
	OR			
Que.3 (a)	What were the Environment-Friendly Practices in Ancient India?	04	CO3	BT2
Que.3 (b)	What are various challenges of traditional medicine in India?	04	CO3	BT2
Que.4 (a)	What are the different initiatives taken by India for preserving its Traditional Knowledge Systems?	03	CO4	BT2
Que.4 (b)	Write a note on Contrast between Traditional and Modern agricultural practices.	04	CO4	BT3
	OR			
Que.5 (a)	"When an elder dies, a library burns", this old African proverb may be the best way to understand traditional knowledge. Explain.	03	CO4	BT2
Que.5 (a)	Explain any four approaches for the conservation and management of nature and Bio-resources.	04	CO4	BT2

Course Outcomes - After Completion of this course student will -

CO1	Explain the origin, evolution and approach of Indian knowledge systems
CO2	Describe importance of Indian Literature and will know about Indian Scholars
CO3	Understand ancient Indian traditions, practices and knowledge for conservation of resources
CO4	Understand and build various practices to conserve and protect rich Indian Knowledge

current status of IKS
- UG & PG programs
- IKS centres under ministry of educ (IKS mission, Nagpur)
- Schools and ~~edu~~
- IIT programs
- used in various field of especially Ayurveda
- Internship programs
- Heritage walks

QUESTION BANK

Unit 01:

1	What is/Define IKS
2	Why do we need IKS?
3	Mention the prime factor of ancient knowledge
4	What is current status of IKS in India? Briefly explain the reason for that.
5	Mention the important Six features of IKS.
6	How IKS has contributed significantly to the advancement of human civilization and culture? Or- What are the achievements of IKS? → Next page

Economic value
Identity
Received Wisdom
Cultural

Six features
Holistic Approach
Natural Remedies
Preservative Care
Cultural Relevance
Inter disciplinary Nature
community based
Ethics (Non-violence)

Need of IKS
→ patriotism / connection
→ environment
→ preservation
→ Economic
→ Nature Resource management

Benefits of IKS

1) The Art of Sacrifice (control)
2) Holistic approach (efficient press)
3) Identity (patriotism)
4) Social harmony & peace (Vasudhaiva kutumbakam)
5) Environmental sustainability (nature) Panchamahabhutas

Achievements.

- invention of zero, algebra, geometry, trigno
- heliocentrism, lunar phases, eclipses, gravity, planetary orbits
- development of Ayurveda, Siddha, Unani systems
- creational value of monuments like temples, forts, palaces, stupas (vastu) and (Shilp)
  - (vastu) → Sci of architecture
  - (Shilp) → Sci and

Typical Examples (IKS)

Focus on steps of the Logic of Syllogism
linear / cyclical time
Direct / Indirect Lemma

How Indian mind and practices are different from West?
How IKS is relevant for the modern world?
What are the benefits of IKS?
Give classification framework (IKS corpus) of IKS
Give some typical examples (paradigm) of IKS in various domains

Unit 02:

Name which ancient Indian institutions set the highest standard of education in India
According to philosophy and literature who is Astika and Nastika?
What are the contributions of Ved Vyas?
Who founded the Vaishesika school of philosophy?
Explain in brief about Binary system given by Acharya Pingala.
Pingala given his Binary system in which book.
Write about Maharshi Parashars contribution in IKS.
Write about Acharya Banabhatta's contribution in IKS.
Write about Acharya Nagarjuna's contribution in IKS.
What Acharya Nagarjuna concluded about truth in Buddhism?
What Nagarjuna concluded about relativity?
What are the contributions of Acharya Panini?
Why Aryabhatta's legacy is truly unmatched and no one can replicate his achieve
What are the astronomical discoveries of Aryabhatta?
What is Acharya Mahavira's birthplace?
Write about Mahaviras Ganita-sara-Sangraha.
What are the contributions of Bodhayan as Mathematician?
What are the significant contributions of Bhaskarachaya-II as Mathematician an
Who wrote Siddhant-Shiromani, a crown of treatises?
Name the Brahmagupta's birthplace.
What term was used by Brahmagupta for gravity?
Who wrote Brahmasputsiddhant?
What is the contribution of Brahamgupta in Mathematician and Astronomy?
What is Varahamihira's 'Brihat Sanhita'?
What is the major contribution of Varahmihira?
Explain about the importance of Charak Sanhita in Ayurveda.
Write about Charaks contribution in the field of medicine.
Write about Shushrutas contribution in the field of medicine.
Write about concept of Ashtanga by Patanjali.
Write Note on God of Ayurveda Dhanwantari.
Why is the Veda called Apaurseya?
Define Veda

Rishi Kanada Kashyapa
Chandahsastra
Kadambari Novels
Generative Grammar / Ashtadhyayi
Mysore / Bihar
Gist of Maths
Bhaskara-II
Bhimal in Rajasthan, India
gurutvakarsanam
Brahmagupta
Optics / Trigno / Combinatoric / magic 4x4 square
Surgical Sci
Yamas / Niyamas / Asana / Pranayam
unauthored or authoriless

Jataka tales

- Vast collection of stories within Buddha literature → journey of Buddha
- Narrate previous life of Buddha (enlightened being → Buddha)
- → Narrate various forms
  - human
  - animal
  - even gods

} This story ...

What is the main content of four Vedas?

Explain four Vedas and the basis of their classification

Rigveda is mainly focussed on what aspect of life and nature?

What is "Atharveda" in which it is mentioned?

What are Upvedas? Name four Upvedas

Write about three gunas/qualities on which type puranas were written by Vyasa

Define Upnishad and its purpose

How many Upnishad are there and what are they?

Define Vedic and Non-vedic school of philosophy (Aastika and Nastika)

Briefly describe the salient features of the Samnkhy, yoga, Nyaya, Vaisheshic, Purva M and vedanga school of philosophy

Unit 03:

In Srimad Bhagavad Gita, 4.33,37-38, What Lord Krishna tells Arjuna about Knowled

Entire body of Organized Knowledge is divided in which 02 ways? Para Vidya & Apara Vidya

Define Traditional Knowledge System as per United Nations

Myths are related to what? Stories having rich source of knowledge, values & cultural identity

What are the three basic psychological components of Hindu Religion? Brahma / Ishwar / Gurus

What is the Moral of the story "Arjuna's Concentration"?

What is the Moral of the story "Prahlada's faith in God"?

Define Rituals.

Write quote on Rituals as given by Swami Vivekanand. Rituals are steps on the Path

How many Rituals are there in Hinduism?

Define Spirituality.

Name Spiritual Guru in Ancient India Valmiki, Vyasa, Adi Shankara, Mahvira

What is the moral of the story "Integrity of Ram"?

What is the moral of the story "Lord Krishna and Gowardhan hill"?

Name the five parts of Panchatantra

Name the five volumes in which Panchatantra is divided into

What is the morale of the story "Monkey and the Crocodile"?

What do you mean by Jataka Tales? Explain in few words.

What is Singhasan Battisi? 32 apsaras / King Bhoj* / King Vikramaditya

Write short note about Sustainable Living Practices in Ancient India.

What were the Environment-Friendly Practices in Ancient India?

Write in short about Ancient Literature on Environment Conservation.

What "Ayurveda" means in Sanskrit?

Write any four benefits of traditional medicine in India.

What are various challenges of traditional medicine in India?

Ancient literature on
Environment conservation

- Atharvaveda (nature harmony) [Vedas]
- Ramayana & Mahabharata (resources) [Epic]
- [Manusmriti] - punishments for harming nature
- [Arthashastra] - sustainable management of natural resources
- Vanaprastha
  - living a simple life in nature

Explain in short about various major traditional medicine systems in India

What are the four Nobel Truths preached by Lord Gautam Buddha?

What is the Mahavira's ultimate message preaching universal compassion?

Spiritualism in India revolves around which three central values?

What are the common practices and rituals that Hindus follows?

Q4:

Indian Traditional knowledge is comprising of?

What is the main objective of Indian Knowledge system?

What is the concept of proposing Cultural Heritage Information System? (CHIS)

Why Nogoya Protocol is introduced?

What Nagoya Protocol proposed?

In which year Biodiversity Act is introduced? 2002

What support is provided by the National Biodiversity Authority?

What are the different initiatives taken by India for preserving its traditional knowledge systems?

Explain various strategies and technological aids for the preservation, management of Indian Knowledge system.

What are the different methods used for the preservation and documentation of Indian knowledge system?

What is the role of Natural Resource Management?

Which are the different areas that encompasses by the Natural Resource Management?

Explain any four approaches for the conservation and management of nature and bio-resources.

Why it is important to document the Indian Knowledge System?

What are several approaches and strategies for the protection and conservation of the Indian knowledge system?

Write a note on Contrasts between Traditional and modern agricultural practices.

Write a note on Contrasts between TK and modern knowledge system.

What are the different benefits of Indian Knowledge System? Explain any four.

Write in short about Traditional Knowledge Digital Library (TKDL).

Write short note on Traditional Knowledge (TK) Digital Database.

BA support)

Regulatory support - permissions & Rules

Financial -

Advisory -

Knowledge Sharing & Collaboration

S/24/R

St. Vincent Pallotti College of Engineering & Technology
(An Autonomous Institute)
B. Tech Fourth Semester Computer Science and Business Systems
End Semester Examination

Course: - Indian Knowledge System
Duration: - 1.30 hours

Course Code: - ES106T
Max. Marks: - 35

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) Solve Que. No. 08 OR Que. No. 09
6) All Questions carry marks as indicated

No.	Description of Question	Mark	CO	BTL
1(a)	What is the goal of Ancient Indian Education?	01	1	2
1(b)	Who founded the Vaisesika school of philosophy?	01	2	1
1(c)	Name the traditional medicine systems in India.	01	3	1
1(d)	In which year Biodiversity Act is introduced?	01	4	1
1(e)	What is the Moral of the story "Arjuna's Concentration"?	01	3	2
2(a)	Give classification framework (IKS corpus) of IKS.	04	1	2
2(b)	Describe prime factors of Ancient Indian Knowledge.	04	1	2
3(a)	Discuss the achievements of IKS.	04	1	2
3(b)	Explain any four features of IKS.	04	1	2
4(a)	Explain the importance of Charak Sanhita in Ayurveda.	04	2	2
4(b)	What are the astronomical discoveries of Maharshi Aryabhatta?	03	2	1
5(a)	Enlist four Vedas. What are the main content of four Vedas?	04	2	1
5(b)	Discuss Contributions of Acharya Nagarjuna in IKS.	03	2	2
6(a)	Describe Sustainable Living Practices in Ancient India.	04	3	2
6(b)	What are the common practices and rituals that people follows?	03	3	2
7(a)	Explain in short about various major traditional medicine systems in India.	04	3	2
(b)	What "Ayurveda" means in Sanskrit? What are various challenges of traditional medicine in India?	03	3	2
(a)	"When an elder dies, a library burns", this old African proverb may be the best way to understand traditional knowledge. Explain.	04	4	2
(b)	What are the different initiatives taken by India for preserving its Traditional Knowledge Systems?	04	4	2
a)	Explain any four approaches for the conservation and management of nature and bio-resources.	04	4	2
b)	Write a note on Contrast between Traditional and modern agricultural practices.	04	4	2

St. Vincent Pallotti
College of Engineering & Technology
(An Autonomous Institute)

Academic Year: 2023-24

CAT I - B.Tech

rse: -	Marketing Research and Marketing Management	**Course Code: -**	CB406T
		Max. Marks: -	20
ester: -	IV	**Duration: -**	1 Hr
gram : -	Computer Science & Business Systems	**Date of Paper**	21/03/2024

ctions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated

ue. No.	**Description of Question**	**Marks**	**[CO]**	**[BTL]**
e.1(a)	What do you mean by Marketing?	02	01	01
e.1(b)	What is Product Life-Cycle?	02	03	01
e.1(c)	How does the product packaging impact the marketing?	01	02	02
e.2	What do you mean by Marketing analysis? How it differs from Marketing Research?	08	01	02
	OR			
e.3	Explain with examples the various stages of PRODUCT DEVELOPMENT.	08	02	02
e.4	Explain the 7 elements of Marketing mix in detail.	07	02	02
	OR			
e.5	How does the consumer behaviour impact the promotion of a product?	07	01	02

e Outcomes - After Completion of this course student will -

01	understand the Concept of Marketing in service sector
02	2. understand the various Marketing Mix and their application
03	3. understand the product life cycle and Branding
04	4. understand Pricing Methods & Price determination Policies and analysis of data
05	understand Market Research and understand various Channels of Marketing

St. Vincent Pallotti
College of Engineering & Technology

Academic Year: 2023-24

CAT II - B.Tech

	Marketing Research and Marketing Management	Course Code: -	CB406T
r: -	IV	Max. Marks: -	20
: -	Computer Science & Business Systems	Duration: -	1 Hr
		Date of Paper	14/05/2024

to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) All Questions carry marks as indicated

	Description of Question	Marks	[CO]	[BTL]
	What do you mean by a promotional mix in marketing?	02	01	01
	What is a survey questionnaire?	02	02	01
	Explain Pricing Research.	02	02	01
	Define Digital Marketing?	02	01	01
	Differentiate between Sale and Marketing?	02	02	02
	Explain how price is determined in a market with the help of examples. Draw chart/graph to justify your answer.	05	01	02
	OR			
	Demonstrate the concept of data analysis.	05	02	02
	What is B2B Marketing? How is it different from Marketing? Explain with examples.	05	02	02
	OR			
	Demonstrate 5 Ms' of Advertising in detail.	05	01	02

tcomes - After Completion of this course student will -

understand the Concept of Marketing in service sector
understand the various Marketing Mix and their application
understand the product life cycle and Branding
understand Pricing Methods & Price determination Policies and analysis of data
understand Market Research and understand various Channels of Marketing

St. Vincent Pallotti College of Engineering & Technology
(An Autonomous Institute)
B. Tech Fourth Semester Computer Science & Business Systems
End Semester Examination

5/24/R

: - Marketing Research and Marketing Management — Course Code: - CB406T

n: - 1 hour 30 min — Max. Marks: - 35

ons to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) All Questions carry marks as indicated

Description of Question	Mark	CO	BTL
at do you mean by Marketing? Give one example.	02	01	02
at do you mean by a decline in product life cycle?	02	02	01
lain in a sentence what is audio visual channel of promotion? Give an example.	02	03	02
at is a Survey Questionnaire?	02	04	01
at do you mean by digital marketing? Give one example	02	05	02
lain Marketing of Services, Importance of marketing in service sector.	05	01	02
at is New Product development & strategy? Also explain the various Stages in New duct development, Product decision.	05	02	02
at is Market Segmentation? Discuss the Basis of segmentation & strategies in ction of segments.	05	01	02
at is Branding & packaging? How does it impact the scope of marketing?	05	02	01
lain how price is determined in a market with help of examples. Draw chart/graph stify your answer.	05	03	02
lain in detail the various types of Market Research. Also discuss the limitations.	05	04	04
t are the different channels of Marketing? Explain in detail any three channels.	05	03	02
t is data analysis? Explain in detail.	05	04	03
rnet Marketing is a powerful device in modern Marketing era." - Discuss in detail.	05	05	05
t is B2B Marketing? How is it different from Marketing? Explain with examples.	05	05	03

St. Vincent Pallotti College of Engineering & Technology

Academic Year: 2023-24

CAT I - B. Tech

urse: -	Operations Research	Course Code: -	CSE351
		Max. Marks: -	20
mester: -	IV	Duration: -	1 Hr.
ogram: -	Computer Science & Business Systems	Date of Paper: -	22/03/2024

Instructions to Candidate - 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03.
3) Solve Que. No. 04 OR Que. No. 05.
4) All Questions carry marks as indicated.
5) Use of Non Programmable calculator is allowed.
6) Use graph paper for related questions.

Q.	Description of Question	Marks	CO	BTL
(a)	Discuss slack and artificial variables.	02	1	2
(b)	Discuss the concept of Linear programming problem.	02	2	2
(c)	Define a project.	01	3	1
(a)	Describe the phases of OR problem solving approach.	04	1	2
(b)	Demonstrate the fundamental theorem of duality.	04	1	3
	OR			
(a)	Solve using simplex method: Maximize $2x1 - x2 + x3$ Subject to $2x1 + x2 - 2x3 \leq 8$ $4x1 - x2 + 2x3 - 2$ $2x1 + 3x2 - x3 \geq 4$ $x1, x2, x3 \geq 0$.	04	1	5
(b)	Illustrate the determination of critical paths	04	3	2
(a)	Demonstrate transportation problem in OR.	04	2	3
(b)	Outline the steps of NWCR method.	03	2	2
	OR			
(a)	Demonstrate Assignment problem in OR.	04	2	3
(b)	Classify various project scheduling techniques.	03	3	4

S-17
IV CSBS
(31)

St. Vincent Pallotti College of Engineering & Technology

(An Autonomous Institute)

Academic Year: 2023-24

CAT II - B. Tech

rse: -	Operations Research	**Course Code: -**	CB405T
		Max. Marks: -	20
ester: -	IV	**Duration: -**	1 Hr.
gram: -	Computer Science & Business Systems	**Date of Paper:-**	15/05/2024

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03.
3) Solve Que. No. 04 OR Que. No. 05.
4) All Questions carry marks as indicated

No.	Description of Question	Marks	[CO]	[BTL]
1(a)	What is an inventory?	02	4	2
1(b)	What is a queue?	02	5	2
1(c)	Define Critical Path.	01	3	1
2(a)	Explain POQ model.	04	4	2
2(b)	Discuss the concept of sensitivity analysis.	04	4	2
	OR			
3 (a)	Enlist the benefits of EOQ model.	04	4	5
3 (b)	Illustrate the process of ABC analysis with example.	04	4	2
4 (a)	Demonstrate the steps of simulation.	04	5	3
4 (b)	Outline any five applications of simulation.	03	5	3
	OR			
5 (a)	Describe the process of random number generator.	04	5	3
5 (b)	Draw a Gantt chart for a project of your choice.	03	3	6

rse Outcomes - After Completion of this course, the student will be able to -

	Recognize the importance and value of Operations Research and linear programming in solving practical problems in industry.
	Interpret the transportation models' solutions and infer solutions to the real-world problems.
	Recognize and solve assignment problems.
	Gain knowledge of drawing project networks for quantitative analysis of projects.
	Know when simulation and dynamic programming can be applied in real world problems.

5/24/P

St. Vincent Pallotti College of Engineering & Technology
(An Autonomous Institute)
B. Tech Fourth Semester Computer Science & Business System
End Semester Examination

Course: - Operations Research
Course Code: - CBA05T
Duration: - 1 hour 30 min
Max. Marks: - 35

Instructions to Candidate – 1) Question No. 1 is compulsory.
2) Solve Que. No. 02 OR Que. No. 03
3) Solve Que. No. 04 OR Que. No. 05
4) Solve Que. No. 06 OR Que. No. 07
5) Solve Que. No. 08 OR Que. No. 09
6) Solve Que. No. 10 OR Que. No. 11
7) All Questions carry marks as indicated
8) Use of Non Programmable calculator is allowed.

Q. No.	Description of Question	Mark	CO	BTL
Q.1(a)	List two types OR problems.	01	1	1
Q.1(b)	What do you mean by Constraints?	01	2	1
Q.1(c)	Discuss what estimation of Project time is.	01	3	1
Q.1(d)	Define Little's Law.	01	4	1
Q.1(e)	Explain is M/M/1 Model.	01	5	1
Q.2	Demonstrate the concept of Linear Programming and explain Linear algebra.	06	1	2
Q.3	Summarize Simplex Algorithm and explain dual-simplex and primal dual algorithms.	06	1	2
Q.4	Infer what is Test of Optimality? Explain MODI method.	06	2	2
Q.5	Compare Balanced & unbalanced Assignment Problems.	06	2	3
Q.6	Elaborate the concept of Critical path and determine its calculation with example.	06	3	4
Q.7	Categorize Gantt chart, PERT & CPM.	06	3	4
Q.8	Outline the creation of Inventory Policy with example.	06	4	5
Q.9	Interpret the concept of Queuing Theory and list its characteristics.	06	4	4
Q.10	Illustrate the concept of Random number generator.	06	5	3
Q.11	Demonstrate examples of Discrete Event System.	06	5	3